(Un)decidability of the Value problem for Two-clock Weighted Timed Games

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FOR SOFTWARE SYSTEMS

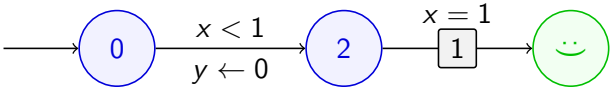
Definition

Weighted Timed Automata

- Clocks $x, y, \ldots$
- Guards $x \bowtie k$ and resets to zero $y \leftarrow 0$
- Weights in $\mathbb{N}$ (or $\mathbb{Z}$) on states & transitions

♣ Motivations

Real-time systems, energy consumption



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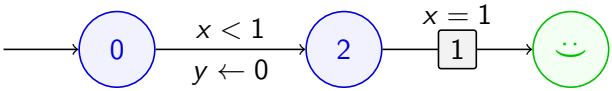
$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$



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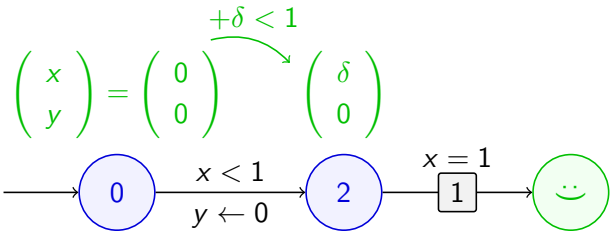

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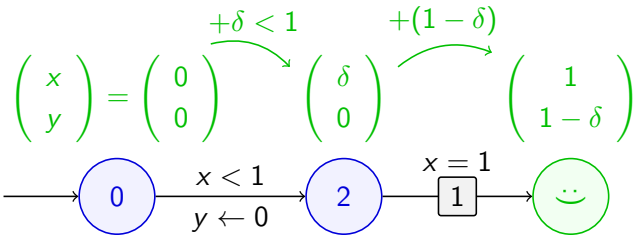

Weight $= (1-\delta) \cdot 2 + 1$

Value $= \inf_{\delta<1}(1-\delta) \cdot 2 + 1 = 1$

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Weighted Timed Games

- States divided between **Min**erva and **Max**imilian
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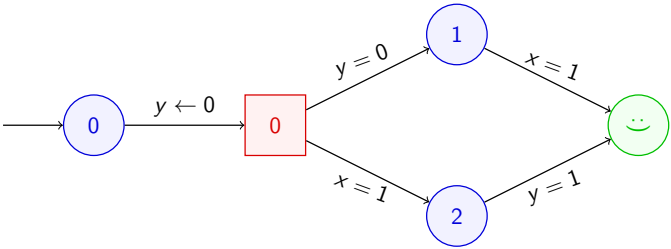
♣ Motivations

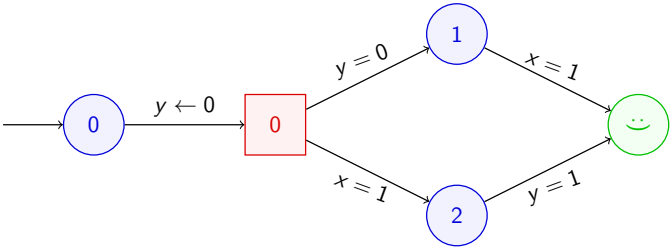
Min: Program,
Max: Environment
(ext. inputs, user)



Definition

Value problem

$$\begin{aligned} Value(\mathcal{G}) &= \inf_{\sigma_{\mathsf{Min}}} \sup_{\sigma_{\mathsf{Max}}} Weight(\sigma_{\mathsf{Min}}, \sigma_{\mathsf{Max}}) \\ &= \sup_{\sigma_{\mathsf{Max}}} \inf_{\sigma_{\mathsf{Min}}} Weight(\sigma_{\mathsf{Min}}, \sigma_{\mathsf{Max}}) \end{aligned}$$



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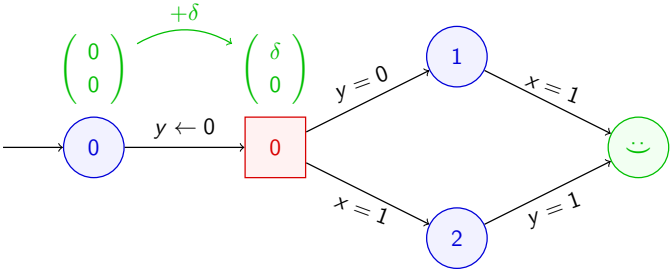

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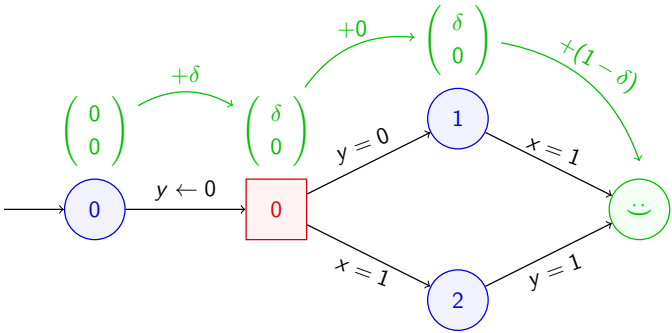

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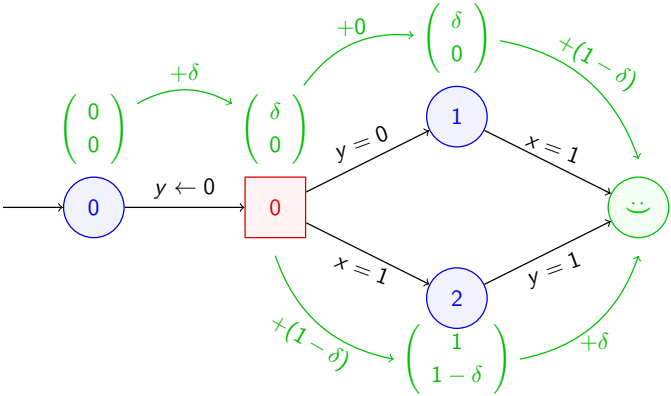

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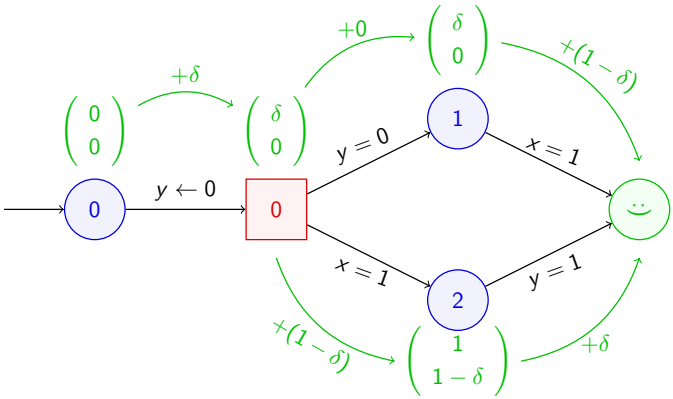
$$\sigma_{\mathsf{Max}} = \max \begin{cases} (1-\delta)\cdot 1 \\ \delta \cdot 2 \end{cases}$$

$$\sigma_{\mathsf{Min}} = \inf_{\delta} \cdots = 1,$$

for $\delta = \frac{1}{3}$



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Number of clocks	Weights in	WTG
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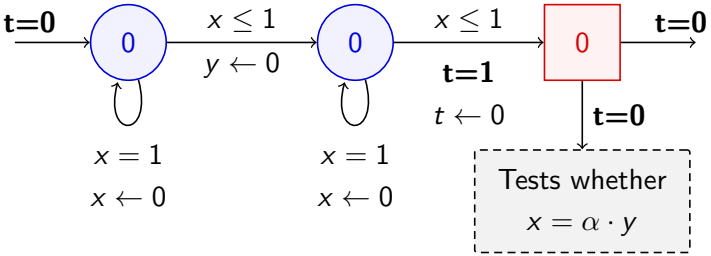
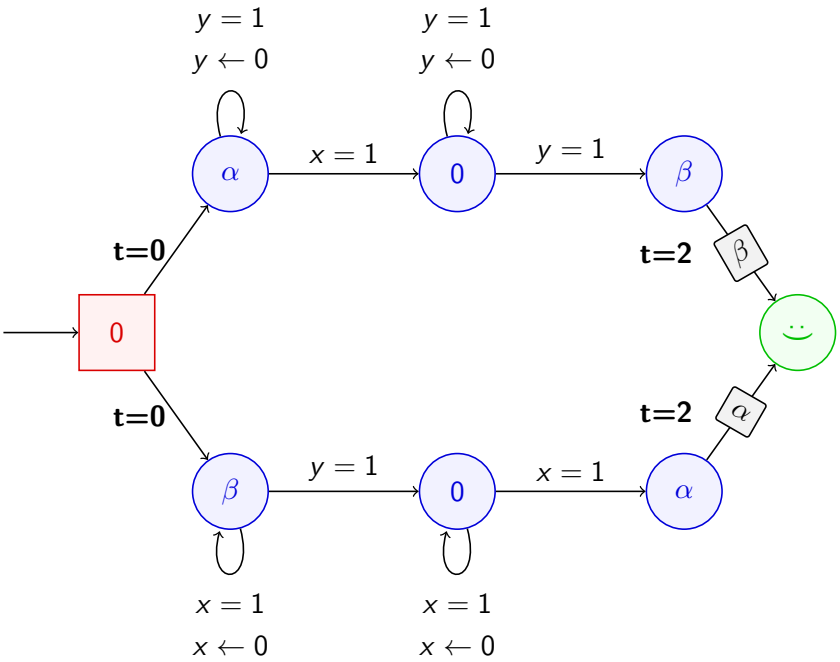
2CM reduces to 3-clock WTG

To simulate $(c, d) \to (c', d')$, $\begin{cases} x = \frac{1}{2^c 3^d} \\ y = \frac{1}{2^{c'} 3^{d'}} \\ t \text{ is a ticking clock} \end{cases}$



Multiplication control module (3 clocks)



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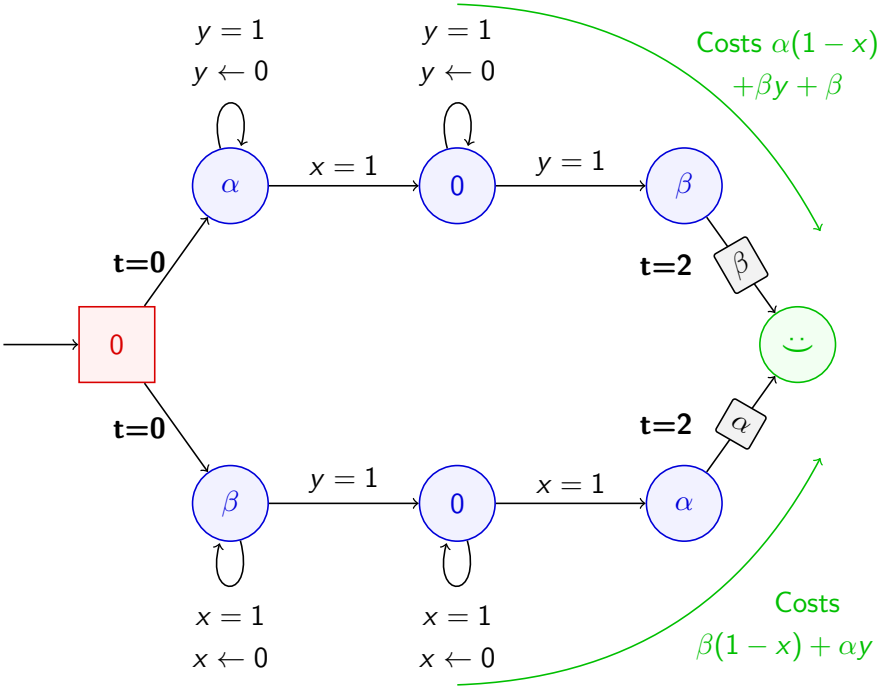

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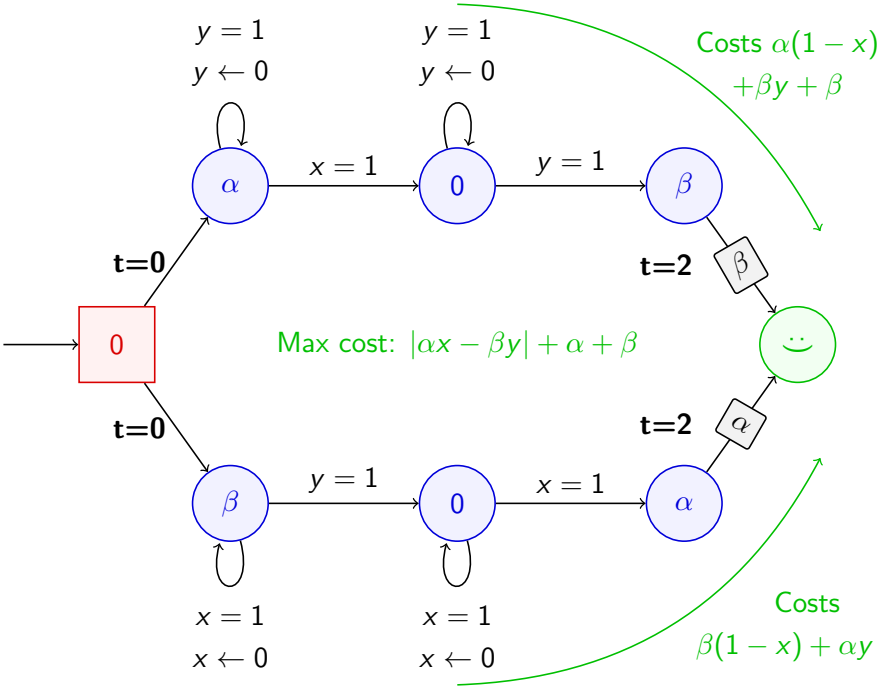

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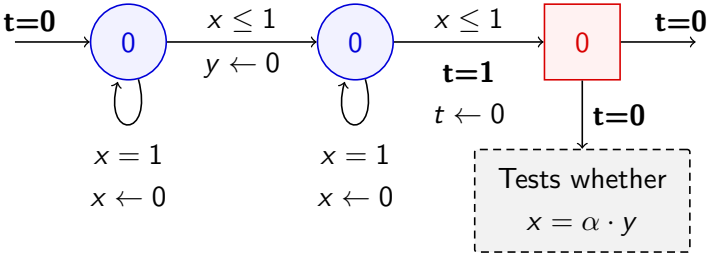

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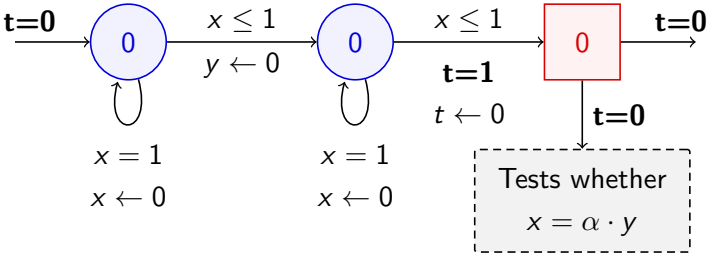

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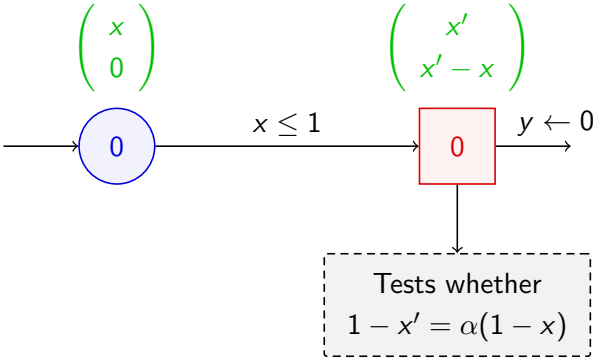
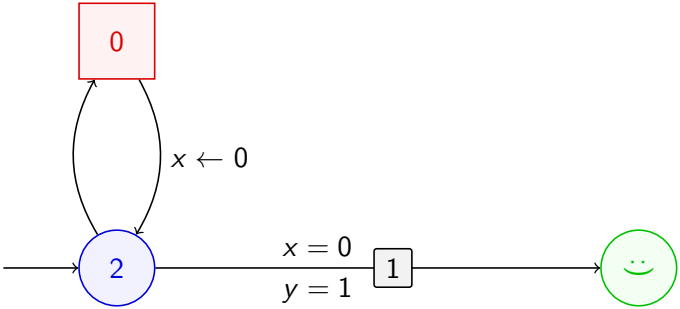


A nifty 2-clock gadget



Value $= 2$

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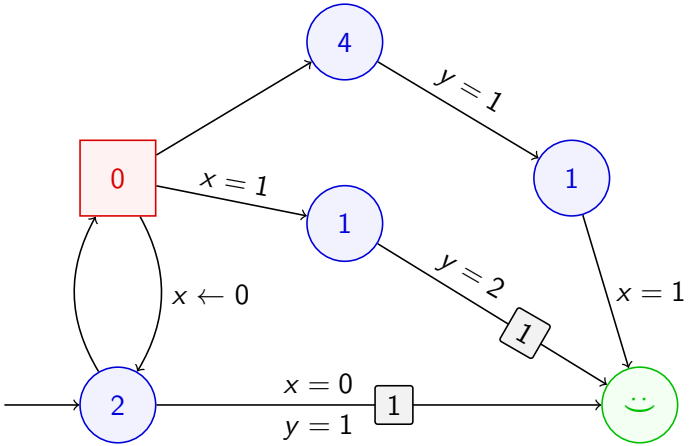

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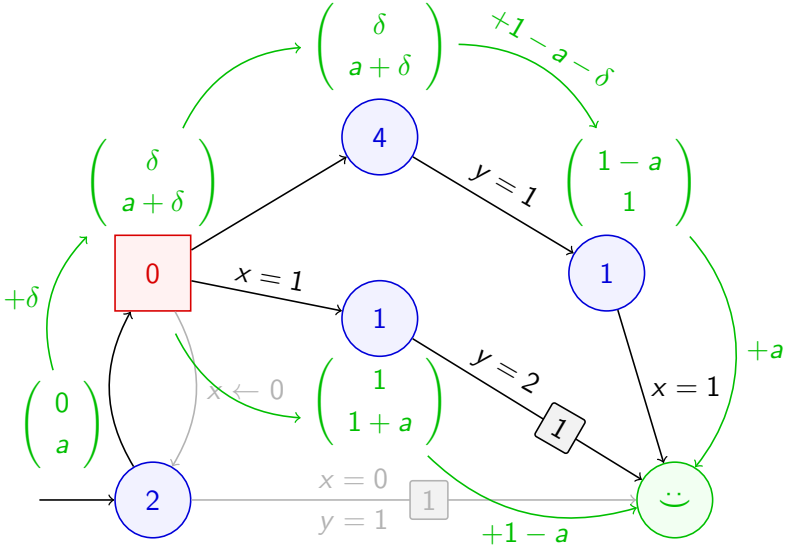

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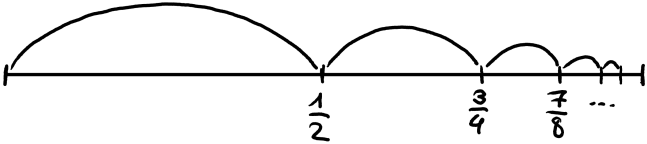
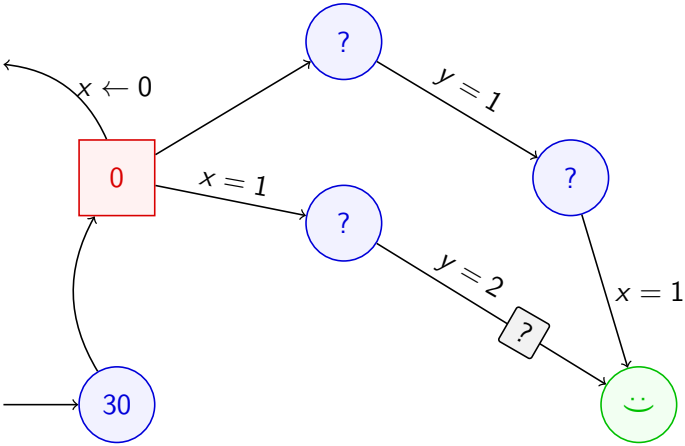


Multiplication control module (2 clocks)

Ensures that $1 - x' = \frac{1}{2}(1 - x) \rightarrow$ Tweak it to ensure $1 - x' = \alpha(1 - x)$



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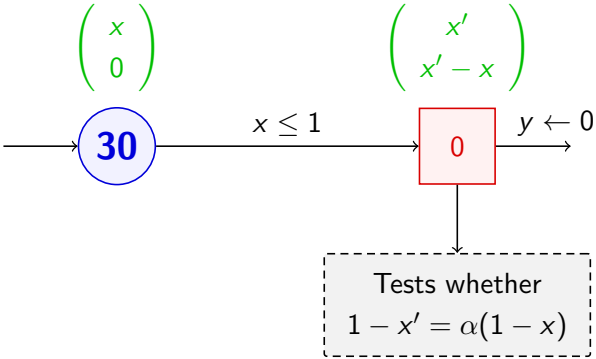

Now what ?

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Let's study a subclass of WTGs!

(Almost) Divergence

Definition (Non-Zeno (or divergent) WTG)

Every cycle has weight ≥ 1

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Every cycle has weight $\geq 1 \Rightarrow$ Value decidable

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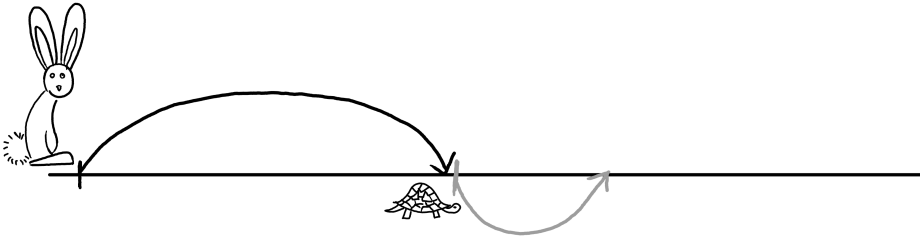

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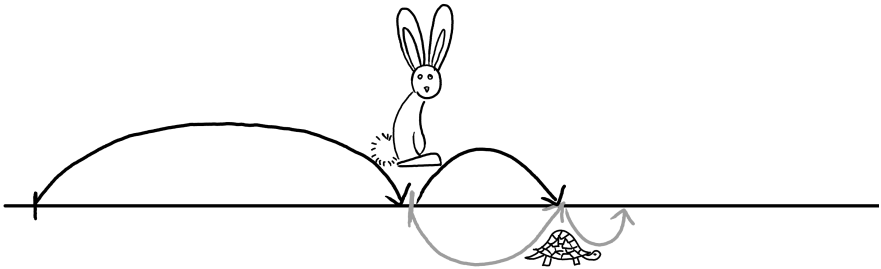

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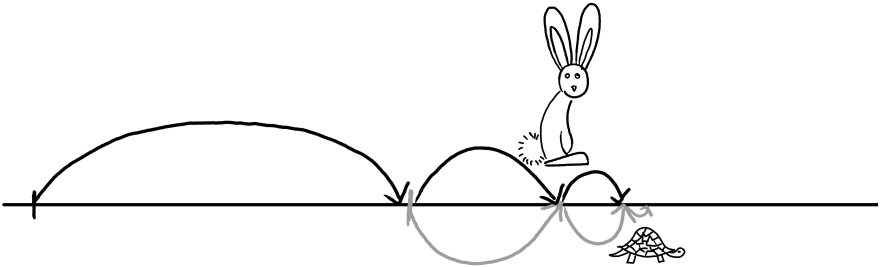

Our 2-clock gadget: very Zeno

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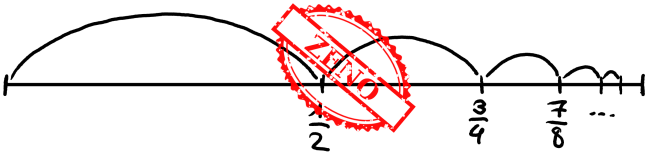

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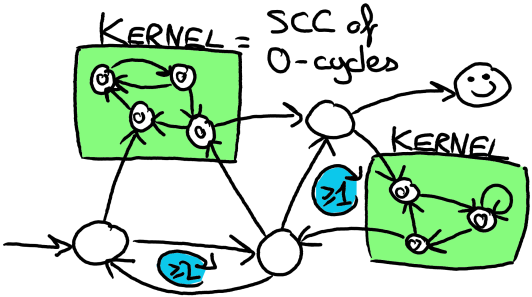

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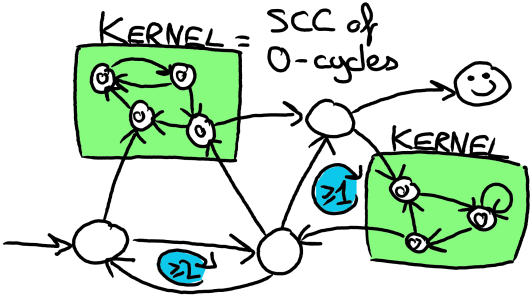
Definition (Almost non-Zeno WTG)

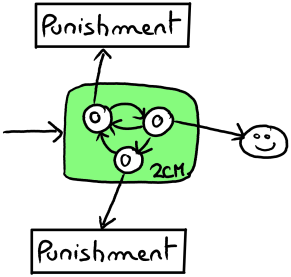
Every cycle has weight $= 0$ or $\geq 1 \;\; \Rightarrow$ Undecidable for ≥ 3 clocks



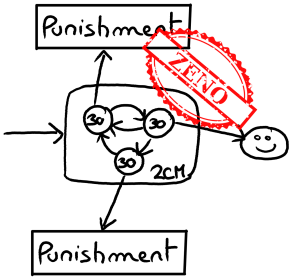
(Almost) Divergence

3-clock reduction

→ Almost non-Zeno



2-clock reduction



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Number of clocks	Weights in	WTG	Almost divergent WTG	Divergent WTG
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(Almost) Divergence: Partial unfolding

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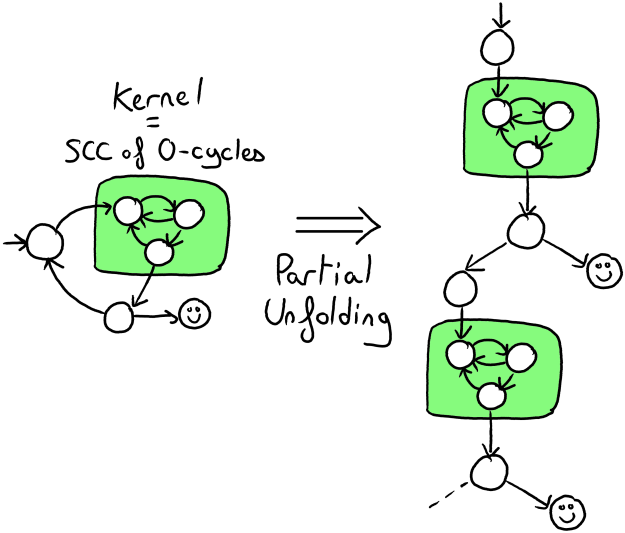

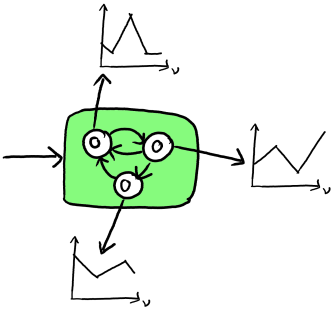

What happens in two-clock kernels?

Value iteration algorithm

Build $\texttt{opt}_k$: set of configurations $\to \mathbb{R}$,

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What happens in two-clock kernels? VIA terminates

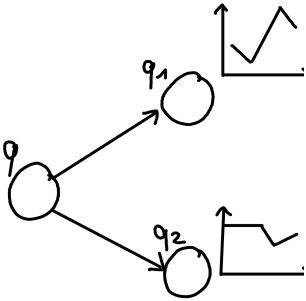
VIA terminates: Proof sketch

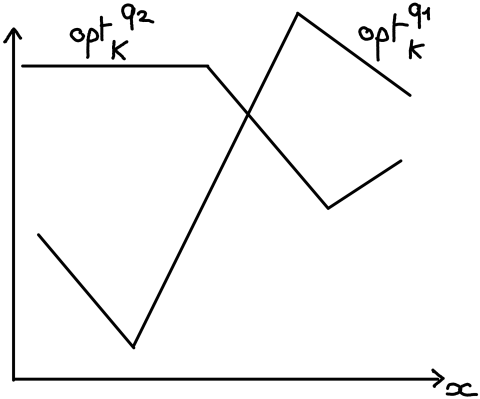
Some "cute" simplifications

- Clock values in $[0, 1[$
- Relaxing guards
- Adding resets *everywhere*

VIA terminates: Proof sketch

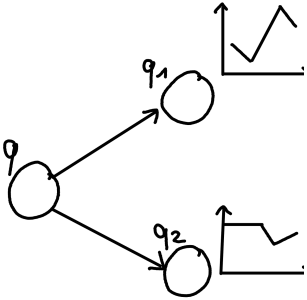
What form can my opt^k functions take ?

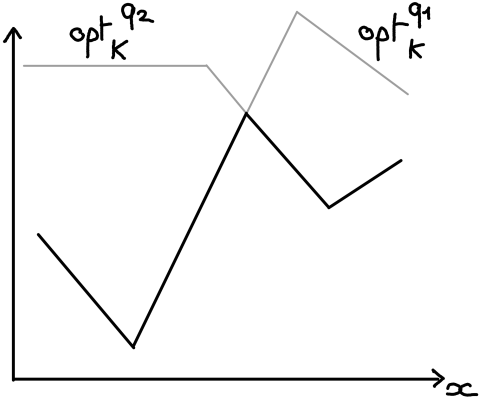




VIA terminates: Proof sketch

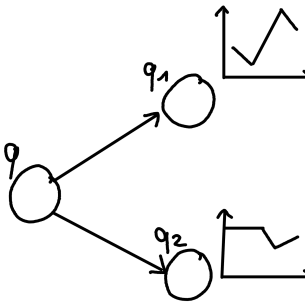
What form can my opt^k functions take ?

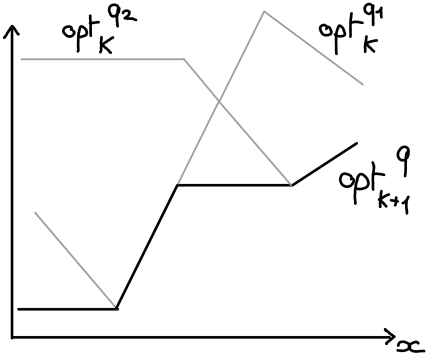




VIA terminates: Proof sketch

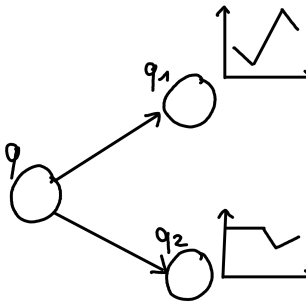
What form can my opt^k functions take ?

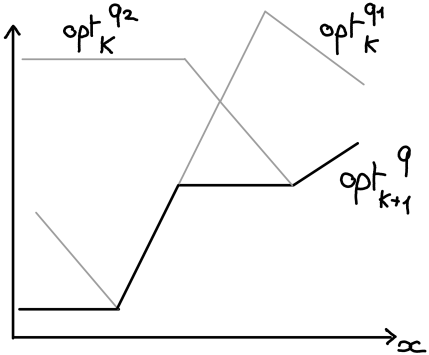




VIA terminates: Proof sketch

What form can my opt^k functions take ?





Finite number of combinations: VIA terminates

Conclusion

Number of clocks	Weights in	WTG	Almost divergent WTG	Divergent WTG
1	$\mathbb{N}$	Decidable[2]	Decidable	Decidable
	$\mathbb{Z}$	Decidable[5]	Decidable	Decidable
2	$\mathbb{N}$	**Undec.**	**Decidable**	Decidable
	$\mathbb{Z}$	**Undec.**	**Decidable***	Decidable
≥ 3	$\mathbb{N}$	Undec.	Undec.[3]	Decidable[1]
	$\mathbb{Z}$	Undec.	Undec.	Decidable[4]

[1]Bouyer et. al. 2005, [2]Bouyer et. al. 2006, [3]Bouyer et. al. 2015, [4] Busatto-Gaston et. al. 2017, [5]Monmege et. al. 2024, * A. Wuttke.

Next big question: Approximability

Number of clocks	Weights in	WTG	Almost divergent WTG	Divergent WTG
1	$\mathbb{N}$	Decidable[2]	Decidable	Decidable
	$\mathbb{Z}$	Decidable[5]	Decidable	Decidable
2	$\mathbb{N}$	**Undec.**	**Decidable**	Decidable
	$\mathbb{Z}$	**Undec.**	**Decidable***	Decidable
≥ 3	$\mathbb{N}$	Undec.	Undec.[3]	Decidable[1]
		???	*(Approximable)*[3]	
	$\mathbb{Z}$	Undec.	Undec.	Decidable[4]
		(Non approx.)[7]	*(Approximable)*[6]	

[1]Bouyer et. al. 2005, [2]Bouyer et. al. 2006, [3]Bouyer et. al. 2015, [4] Busatto-Gaston et. al. 2017, [5]Monmege et. al. 2024, [6]Busatto-Gaston et. al. 2018, [7]Guilmant & Oauknine 2024, *A. Wuttke.

Thank you
for listening!